

Procurement review before Gerber release. Net names are frozen per the V1.1 design document.

Charge policy (V1.3, review list 2.2 / 2.3)

Power input definition (V1.6): compatible with a 5 V / 2 A source; continuous 2 A at elevated ambient is NOT guaranteed.

F1-F3 PTC hold current derates with temperature – verify at 25 / 40 / 50 °C before claiming 2 A.

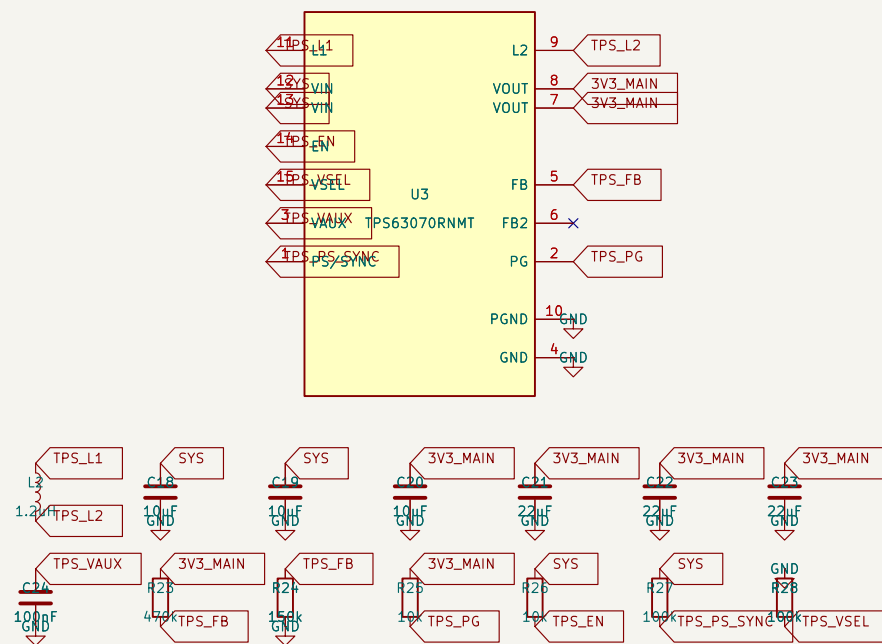
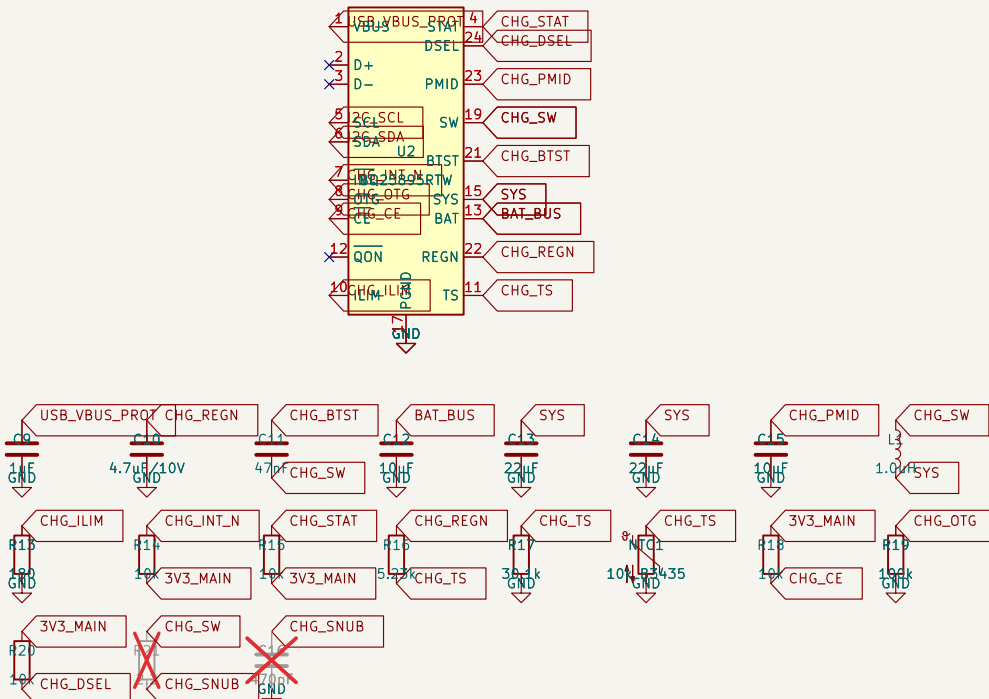
default = 896 mA (14 x 64 mA) – keeps the Molex 53261–0271 connector, 26 AWG hangers and terminals inside their thermal budget.

/CE = CHG_CE (GPIO47), 10k pull-up to V3G_MAIN –> charge DISABLED at power-up.

up: MCU boot –> I2C init –> write ICHG / INILIM / ITERM / VREG –> read back and verify –> drive /CE LOW –> charging enabled.

Bench test at peak load (Wi-Fi TX + EPD refresh + TF card + CPU) required: measure J4/J5 peak current and connector / terminal / hanger temperature rise; limit concurrent load in firmware if the design target is exceeded.

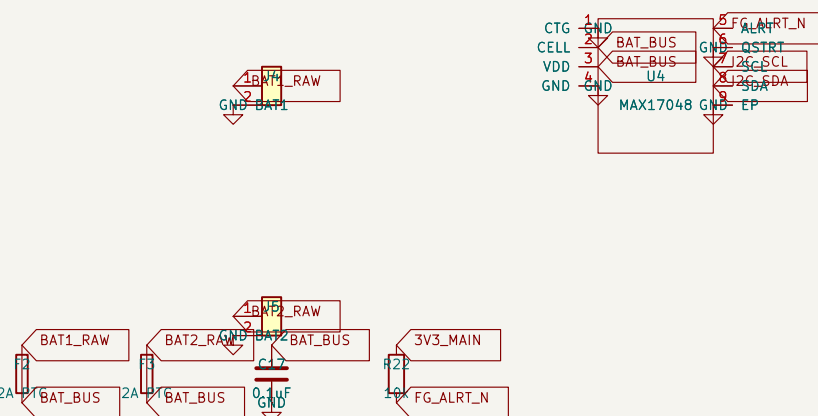
05 TPS63070 3V3 buck-boost (VOUT8 = Passive, no power-output clash)



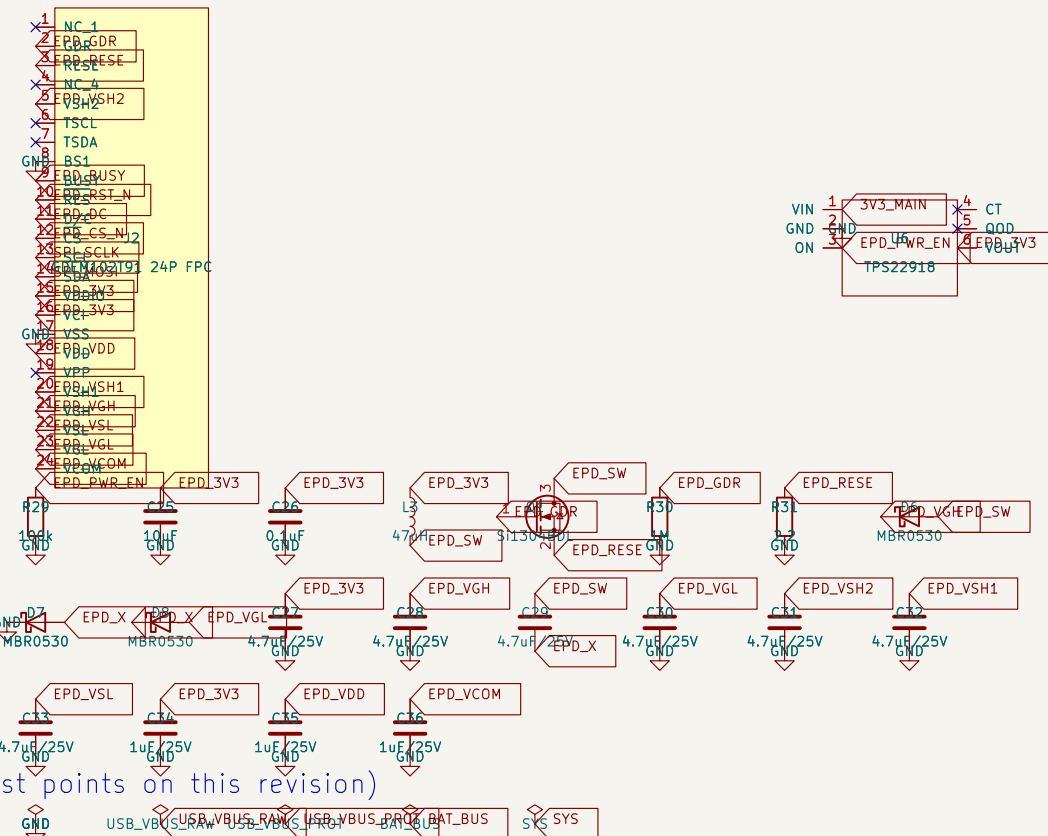
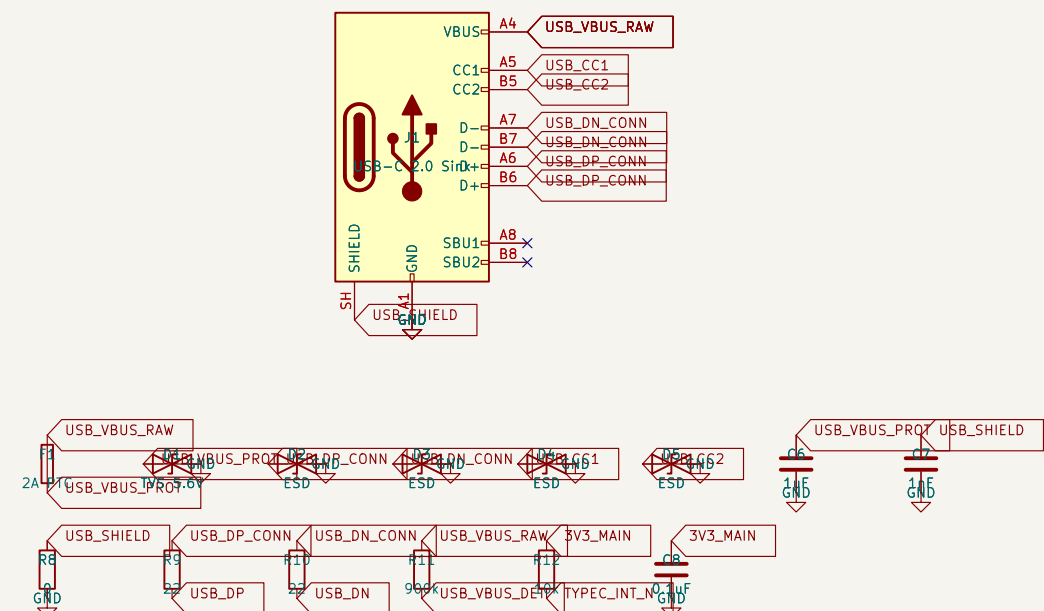
04 Battery inputs (either or both) + MAX17048 fuel gauge

06 EPD GDEM102T91 24P FPC + SSD1677 HV network (locked to panel typical application circuit)

CC1	4	USB_CCI	7	206DA
CC2	2	USB_CCI2	5	9CCCL
PORT	4	USB0USB0	5	0
VBUS_DET	3	USB0USB0	4	GND
ADDR	4	USB320L1	5	GN_N
INT_N	5	TYPEC_INT_N	5	V50MAIN



07 MicroSD (SPI, shared SCLK/MOSI with EPD, independent CS)



08 PWR_FLAG (no test points on this revision)

